The Rank Price:

the interference dimension of Clifford+T circuits

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Abstract

Representations of a Clifford+T phase polynomial modulo Clifford operations form a coset of the punctured Reed–Muller code $\mathrm{RM}(h-4,h)^*$. The optimization literature decodes this coset for *weight*: the number of odd-coefficient parities is the T-count, and its exact minimization is NP-hard. We decode the same coset for *rank*: the $\mathbb{F}_2$-dimension δ^* of the span of the odd-parity support, minimized over the coset, which we call the *interference dimension*, because exact path-sum simulation costs at most $\mathrm{poly}\cdot 2^{\delta^*}$ rather than 2^{count}. We prove that the subspaces supporting some coset element form an intersection-closed family, so a unique minimal *core* subspace V_0 exists with $\delta^* = \dim V_0$; that the Clifford-equivalence class is computed by a syndrome map into the shortened code $\mathrm{RM}(3,h)$; and that δ^* equals the $\mathbb{F}_2$-rank of an explicit $h \times O(h^2)$ matrix read off the phase polynomial: polynomial time, in sharp contrast with the NP-hardness of the weight objective. For the standard $\mathrm{GF}(2^k)$ multiplier benchmarks the invariant is a convolution tensor and $\delta^* = 4k-1$ exactly; combined with the hierarchy $\log_2(t^*+1) \le \delta^* \le t^*$ this yields a linear lower bound on multiplier T-count in the coset model, which meets bilinear-complexity upper bounds at $t^* = \Theta(k)$. A counting argument gives families on which the rank price beats the optimal weight price superpolynomially. We validate every theorem by machine (exhaustive small-case checks, 142/142 agreement with brute-force coset scans) and benchmark against the `feynman` suite and the `quizx` stabilizer decomposer, with both engines agreeing on every exact amplitude. Optimal T-count is NP-hard; the interference dimension of a phase polynomial is a matrix rank. The invariant is defined for a fixed set of path variables; it is an upper bound on simulation cost, not an operator invariant.

1 Introduction

Classical simulation of Clifford+T circuits is priced, throughout the literature, by *counts*: the number t of T gates governs stabilizer-rank methods at $2^{\alpha t}$ with $\alpha \le 0.3963$ [BG16, QPG21, KvdW22, KvdWV22], and a long line of compilers minimizes t itself [AM19, HC19, Van24, dBBW20]. Amy and Mosca identified the structure underneath: representations of a $\{\mathrm{CNOT}, T\}$ phase polynomial that implement the same unitary differ by codewords of the punctured Reed–Muller code $\mathrm{RM}(n-4,n)^*$, and T-count minimization is minimum-*weight* decoding of this coset [AM19]; exact weight decoding is NP-hard [vdWA23].

This paper starts from an observation about what a path-sum simulator pays. A simulator that eliminates the Clifford part in polynomial time [Amy19] and enumerates the residual interference coordinates pays 2^{ρ} where ρ is the *rank* of the odd-parity support (the dimension of the $\mathbb{F}_2$-span of the parities carrying odd coefficients), not their number. Two representations in the same coset can have very different ranks, so the natural objective is

$$\delta^* \;=\; \min_{c' \in c + \mathrm{RM}(h-4,h)^*} \dim \mathbb{F}_2\text{-}\operatorname{span}\bigl(\operatorname{supp} c'\bigr),$$

the *interference dimension*. To our knowledge (Section 8) this objective has not previously been formulated: the compiler literature minimizes weight only, and the closest simulation-cost parameters of this shape are properties of a particular circuit trace rather than invariants of the Clifford coset [LC26, CL26a].

Contributions.

1. **Structure** (Section 3). The Clifford-equivalence class of a phase polynomial is computed by a *syndrome map* Φ into the space of no-constant cubics, which is exactly the shortening of $\mathrm{RM}(3,h)$ at 0, dual to $\mathrm{RM}(h-4,h)^*$ (Lemmas 4–5). The vectors $Q(m)$ on ≤ 3-fold combinations of independent vectors are linearly independent (Theorem 6), ≥ 4-fold combinations reduce by a one-line degree argument that re-derives the spider-nest identity of [dBBW20] (Lemma 7), and consequently $W_{V_1} \cap W_{V_2} = W_{V_1 \cap V_2}$ (Theorem 9): the family of supporting subspaces is intersection-closed, a unique minimal *core* V_0 exists, and $\delta^* = \dim V_0$ (Corollary 10).

2. **Algorithm** (Section 4). $V_0^\perp$ is the left kernel of an explicit $h \times (1 + h + \binom{h}{2})$ matrix M_P over $\mathbb{F}_2$ built from the coefficients of the cubic invariant, and $\delta^* = \operatorname{rank} M_P$, in $O(h^4)$ arithmetic and fully constructively (Theorem 13). The weight objective on the same coset is NP-hard [vdWA23]; the rank objective is a matrix rank.

3. **The $4k-1$ law** (Section 5). For the standard $\mathrm{GF}(2^k)$ Mastrovito multiplier circuits the cubic invariant is the convolution tensor $\sum_{i,j} x_i y_j z_{i+j}$ and $\delta^* = 4k-1$ exactly (Theorem 15). With the hierarchy $\log_2(t^*+1) \leq \delta^* \leq t^*$ (Proposition 3) this gives a *linear lower bound* $t^* \geq 4k-1$ on multiplier T-count in the coset model; bilinear-complexity decompositions are legal coset moves, so $t^* \leq 7\, M_{\mathrm{bil}}(k) = O(k)$ and $t^*(\text{multiplier}) = \Theta(k)$: a bridge between T-count optimization and the bilinear complexity of multiplication.

4. **Separation and practice** (Sections 6–7). A counting argument shows generic dense cores on d variables have $\delta^* = d$ but $t^* = \Theta(d^2)$, so the rank price $2^{\delta^*} = 2^{\Theta(\sqrt{t^*})}$ beats the optimal weight price $2^{0.396 t^*}$ superpolynomially. On the `feynman` benchmark suite the rank price beats the generic weight bar on 33/40 circuits, and coset minimization lowers δ^* strictly below the raw parity rank on 26/40. An exact $\mathbb{Z}[\omega_8]$ implementation, cross-validated amplitude-by-amplitude against the `quizx` cat-state decomposer on a counting task with a closed-form answer, exhibits a provably flat $16\times$ cost ratio per k against an empirically accelerating $27/52/65\times$ for the T-count-driven method.

All proofs are elementary ($\mathbb{F}_2$ linear algebra plus classical facts about Reed–Muller codes), and every step is machine-verified (Appendix A).

2 Preliminaries

Throughout $h \geq 4$; $\mathbb{F}_2$ is the two-element field; $\omega = e^{i\pi/4}$. For a nonzero mask $m \in \mathbb{F}_2^h$ and $z \in \mathbb{F}_2^h$ write $\langle m, z\rangle = \bigoplus_{i \in \operatorname{supp} m} z_i$. A *phase function* on h path variables is

$$f(z) \;=\; \sum_S a_S \,\langle S, z\rangle \bmod 8, \qquad a_S \in \mathbb{Z}_8,$$

acting as the diagonal phase $\omega^{f(z)}$ in a sum-over-paths [DHH+05, Amy19]. Each odd-coefficient parity costs one T gate; Clifford diagonal adjustments are exactly the even-coefficient modifications. The *odd pattern* $c(f) \subseteq \mathbb{F}_2^h \setminus \{0\}$ is the set of parities with odd coefficient, identified with a vector in $\mathbb{F}_2^{2^h-1}$ indexed by nonzero points.

$\mathcal{C}_h$ denotes the space of polynomials over $\mathbb{F}_2$ of degree ≤ 3 with zero constant term; $f(h) := \dim \mathcal{C}_h = h + \binom{h}{2} + \binom{h}{3}$. For a nonzero mask m,

$$Q(m) \;:=\; e_1(\operatorname{supp} m) + e_2(\operatorname{supp} m) + e_3(\operatorname{supp} m) \;\in\; \mathcal{C}_h,$$

the elementary symmetric sums of degree ≤ 3 on the support, and the *grading map* is $\Phi(c) := \sum_{y \in c} Q(y)$. The map Q arises from the integer expansion $x_1 \oplus \cdots \oplus x_j = \sum x_i - 2\sum_{i<i'} x_i x_{i'} + 4\sum x_i x_{i'} x_{i''} - \cdots$, whose coefficients of degree ≥ 4 vanish modulo 8: reducing the three grades modulo 2 (a_S mod 2; pair coefficients divided by 2; triple coefficients divided by 4) shows that f modulo Clifford is exactly a cubic in $\mathcal{C}_h$, and a parity maps to Q of its support. (Verified exhaustively at $h = 3$: the 8^7 coefficient vectors reach exactly $8^3 \cdot 4^3 \cdot 2$ distinct functions and the kernel is constant on the grading.)

$\mathrm{RM}(r, m)$ is the Reed–Muller code of evaluations of degree-$\le r$ polynomials on $\mathbb{F}_2^m$; $\dim \mathrm{RM}(r, m) = \sum_{i \le r} \binom{m}{i}$, minimum distance 2^{m-r}, and $\mathrm{RM}(r, m)^\perp = \mathrm{RM}(m - r - 1, m)$ [MS77]. The superscript * denotes puncturing at the point 0.

Two representations on the same h path variables induce the same phase function up to Clifford phases iff their coefficient vectors differ by a relation; Amy–Mosca proved the mod-2 residues of relations form $\mathrm{RM}(h-4, h)^*$ [AM19]. Our development re-derives this self-containedly (Lemma 5); we also note that the naive 0/1 lift of a codeword is *not* itself a relation for $h \ge 5$: an exact lift requires a triangular even-part correction (fix pair grades with S-type coefficients, then linear grades), which we verify constructively for $h \le 6$.

The two decoding problems on the coset $c + \mathrm{RM}(h-4, h)^*$ are

$$t^*(c) = \min_{c'} |c'| \qquad \text{(weight; the } T\text{-count [AM19], NP-hard [vdWA23])},$$

$$\delta^*(c) = \min_{c'} \dim \mathbb{F}_2\text{-span}(\operatorname{supp} c') \qquad \text{(rank; this paper)}.$$

Definition 1. $W_V := \mathrm{span}_{\mathbb{F}_2}\{Q(m) : m \in V \setminus \{0\}\} \subseteq \mathcal{C}_h$ for a subspace $V \le \mathbb{F}_2^h$.

Theorem 2 (Simulation pays at most rank)**.** *Given a representation whose odd parities span a ρ-dimensional space, a computational-basis path-sum amplitude is computable exactly in* $\mathrm{poly}(h) \cdot 2^\rho$ *arithmetic operations in $\mathbb{Z}[\omega]$: rotate the span to the first ρ coordinates, enumerate them, and evaluate the remaining all-even (Clifford) phase by the standard non-branching quadratic Gauss-sum elimination [Amy19]. Hence* $\mathrm{poly}(h) \cdot 2^{\delta^*}$ *operations suffice for any representation in the coset, after the polynomial-time preprocessing of Corollary 14.*

Proof. Apply a $GL(h, \mathbb{F}_2)$ change of variables mapping the odd span to the leading ρ coordinates. Every parity involving a trailing coordinate then has even coefficient. Fix the leading coordinates. A term $2\langle S, z\rangle$ equals $2\sum_{i \in S} z_i - 4\sum_{i<i'} z_i z_{i'}$ modulo 8, and a term $4\langle S, z\rangle$ equals $4\sum_{i \in S} z_i$, so the remaining phase is $\omega^{2q(z)} = i^{q(z)}$ for a $\mathbb{Z}_4$-valued quadratic form q in the trailing variables, with coefficients depending on the fixed leading values. Summing i^q over one trailing variable either yields a scalar in $\{2, 1 \pm i\}$ times $i^{q'}$ for a quadratic form q' in the remaining variables, or imposes one affine constraint that is solved by substitution; neither case branches. Each of the 2^ρ leading assignments therefore costs $\mathrm{poly}(h)$. Our implementation validates exact equality with brute force (Appendix A). □

Remark (scope). δ^* is an invariant of the phase polynomial under Clifford coset moves on a *fixed* set of h path variables. Path-sum rewrites that change h, such as the Hadamard rule $\sum_y (-1)^{y(u+v)} = 2[u = v]$ and nonlinear substitution [Amy19], lie outside the model, so δ^* is not an invariant of the operator. Two circuits for the identity show this: $T\,H\,H\,T^\dagger$ has 3 path variables and $\delta^* = 2$, and Toffoli · Toffoli (with Toffoli $= H \cdot \mathrm{CCZ} \cdot H$) has 7 path variables and $\delta^* = 3$, while the operator has cost 1. Theorem 2 is thus an upper bound; we prove no lower bound on simulation cost in terms of δ^*.

Proposition 3 (Hierarchy)**.** $\log_2(t^*+1) \le \delta^* \le t^*$*, and both ends are achieved.*

Proof. Right: a minimum-weight element spans at most t^* dimensions; a single T gate has $t^* = \delta^* = 1$. Left: a rank-δ^* element lives on δ^* dimensions, where at most $2^{\delta^*}-1$ parities exist, so some representation has $t^* \le 2^{\delta^*}-1$ odd parities. The CCZ pattern achieves equality: its seven parities fill a 3-dimensional space, so $\delta^* = 3$, and $t^* = 7 = 2^3-1$ because every nonzero codeword of $\mathrm{RM}(h-4,h)^*$ has weight at least 15, so every other coset element has weight at least $15-7=8$. (Selinger's 4-T gate [Sel13] is $\mathrm{CCZ}\cdot\mathrm{CS}^\dagger$: its syndrome is $xyz+xy$, a different coset.) Generic dense cores (Section 6) have $t^* = \Theta((\delta^*)^2)$. □

3 The syndrome map and the canonical core

Lemma 4 (Φ is a syndrome map)**.** *For $1 \le |T| \le 3$, the coefficient of x^T in $\Phi(c)$ equals $\#\{y \in c : \operatorname{supp} y \supseteq T\} \bmod 2 = \langle c, \mathrm{ev}(x^T)\rangle$, where $\mathrm{ev}(g) = (g(y))_{y\neq 0}$. Hence $\Phi(c) = 0$ iff $c \perp \mathrm{ev}(\mathcal{C}_h)$.*

Proof. $e_d(\operatorname{supp} y)$ contains x^T iff $T \subseteq \operatorname{supp} y$; and $x^T(y) = [T \subseteq \operatorname{supp} y]$. Sum over $y \in c$. □

Lemma 5 (Exact dual pair)**.** $\mathrm{ev}(\mathcal{C}_h)$ *is the shortening of* $\mathrm{RM}(3,h)$ *at* 0*, of dimension $f(h)$, and $\{c : \Phi(c) = 0\} = \mathrm{RM}(h-4,h)^*$.*

Proof. Orthogonality: for $\deg p \le h-4$ and $g \in \mathcal{C}_h$, $\deg(pg) \le h-1$ and the only multilinear monomial with odd sum over $\mathbb{F}_2^h$ is $x_1\cdots x_h$, so $\sum_{y\neq 0} p(y)g(y) = \sum_y (pg)(y) - p(0)g(0) = 0$. Injectivity of ev on $\mathcal{C}_h$: a nonzero multilinear polynomial vanishing at all $y \neq 0$ is a multiple of the point indicator δ_0, of degree $h > 3$. Dimensions: $\dim \mathrm{RM}(h-4,h)^* = 2^h - 1 - f(h)$ (puncturing preserves dimension since $d_{\min} = 16 > 1$), and the two dimensions sum to the length 2^h-1; exact annihilators. Lemma 4 identifies the kernel of Φ with the annihilator of $\mathrm{ev}(\mathcal{C}_h)$. □

Thus $\Phi(c) = \Phi(c')$ iff $c + c' \in \mathrm{RM}(h-4,h)^*$: Clifford classes are cosets, and Φ computes the syndrome. Weight decoding of this coset is Amy–Mosca; rank decoding is what follows.

Theorem 6 (Independence)**.** *Let $e_1,\dots,e_n \in \mathbb{F}_2^h$ be linearly independent and, for $S \subseteq [n]$ with $1 \le |S| \le 3$, put $m_S := \bigoplus_{i\in S} e_i$. Then $\{Q(m_S)\}_S$ is linearly independent in $\mathcal{C}_h$.*

Proof. Suppose $\sum_{S\in F} Q(m_S) = 0$ with $F \neq \emptyset$. The m_S are distinct and nonzero, so $c_F = \{m_S : S \in F\}$ is a pattern with $\Phi(c_F) = 0$; by Lemma 4, $c_F \perp \mathrm{ev}(\mathcal{C}_h)$. Choose dual functionals λ_j with $\lambda_j \cdot e_i = \delta_{ij}$ and set $g_T := \prod_{j\in T}(\lambda_j \cdot x) \in \mathcal{C}_h$ for $1 \le |T| \le 3$; then $g_T(m_S) = [T \subseteq S]$. Let $S_0 \in F$ have maximal cardinality: $\langle c_F, \mathrm{ev}(g_{S_0})\rangle = \#\{S \in F : S \supseteq S_0\} = 1$, contradicting orthogonality. □

Lemma 7 (Reduction of ≥ 4-fold combinations)**.** *W_V is spanned by the $Q(m_S)$ with $|S| \le 3$, for any basis of V.*

Proof. For $|S| = k \ge 4$ pick distinct $a,b,c \in S$ and set $u_1 = \bigoplus_{i\in S\setminus\{a,b,c\}} e_i$, $u_2 = e_a$, $u_3 = e_b$, $u_4 = e_c$: four independent vectors spanning W. Any $g \in \mathcal{C}_h$ pulls back to a degree-≤ 3 polynomial on $\mathbb{F}_2^4$, whose sum over $\mathbb{F}_2^4$ vanishes; so $\sum_{y\in W\setminus 0} g(y) = -g(0) = 0$ and, by Lemma 4, Φ of the 15-pattern of W is zero. This rewrites $Q(m_S)$ as the sum of Q over the other 14 combinations; every combination containing u_1 is a combination of at most $k-1$ of the e_i, and every combination without u_1 is a ≤ 3-fold; induct. (This one-line degree argument recovers the cubic-level content of the spider-nest identities [dBBW20].) □

Corollary 8. $\dim W_V = f(\dim V)$, *with basis* $\{Q(m_S) : 1 \le |S| \le 3\}$.

Theorem 9 (Intersection lemma)**.** $W_{V_1} \cap W_{V_2} = W_{V_1 \cap V_2}$ *for all subspaces* $V_1, V_2 \le \mathbb{F}_2^h$.

Proof. $\supseteq$ is clear. Take an adapted basis: $c_1, \dots, c_r$ of $V_1 \cap V_2$, extended by $a_1, \dots, a_p$ to V_1 and by $b_1, \dots, b_q$ to V_2; the union is independent. By Corollary 8 the generators of W_{V_1} and W_{V_2} are subfamilies of $\{Q(m_S)\}$ over the union basis, independent by Theorem 6; the families overlap in exactly the $f(r)$ generators indexed by subsets of the c-indices. Hence $\dim(W_{V_1} + W_{V_2}) = f(r+p) + f(r+q) - f(r)$ and $\dim(W_{V_1} \cap W_{V_2}) = f(r) = \dim W_{V_1 \cap V_2}$. □

Corollary 10 (Canonical core)**.** *Let* $\mathcal{V}(c)$ *be the set of subspaces* V *such that the coset* $c + \mathrm{RM}(h-4, h)^*$ *contains a pattern supported in* V*. Then* $V \in \mathcal{V}(c)$ *iff* $\Phi(c) \in W_V$*; the family is closed under intersection; it has a unique minimal element* $V_0(c)$*; and* $\delta^*(c) = \dim V_0(c)$*, with* $\{w :$ *some coset element avoids direction* $w\} = V_0(c)^\perp$.

4 The interference dimension is a matrix rank

Lemma 11 (Annihilator of W_V)**.** *Identify a functional* φ *on* $\mathcal{C}_h$ *with the polynomial* $\hat{\varphi}(x) = \sum_T \varphi_T x^T$*. Then* $\varphi(Q(m)) = \hat{\varphi}(m)$*, and* $\mathrm{Ann}(W_V) = \{\varphi : \hat{\varphi}|_V = 0\}$ *exactly (restriction of no-constant cubics to* V *is surjective, and dimensions match).*

Lemma 12 (Hyperplane cubics)**.** *For* $w \neq 0$*, the no-constant cubics vanishing on* $w^\perp$ *are exactly* $\{\mathrm{ml}(\ell_w \cdot q) : \deg q \le 2\}$*, where* $\ell_w(x) = w \cdot x$ *and* ml *denotes multilinear reduction.*

Proof. Inclusion is clear. The kernel of $q \mapsto \mathrm{ml}(\ell_w q)$ consists of the degree-≤ 2 polynomials vanishing on the affine hyperplane $\{\ell_w = 1\}$ and has dimension h, so the image has dimension $1 + h + \binom{h}{2} - h = 1 + (h-1) + \binom{h-1}{2}$, which equals the dimension of the target space; the inclusion is an equality. □

Theorem 13 (Polynomial-time rank price)**.** *Define* $M_P \in \mathbb{F}_2^{h \times (1+h+\binom{h}{2})}$ *with rows* $i \in [h]$*, columns* $q \in \{1\} \cup \{x_j\} \cup \{x_j x_k\}_{j<k}$*, and entry the coefficient in* $P = \Phi(c)$ *of* $\mathrm{ml}(x_i \cdot q)$*:*

$$M_P[i, 1] = P_{x_i}, \qquad M_P[i, x_j] = \begin{cases} P_{x_i} & j = i \\ P_{x_i x_j} & j \neq i \end{cases}, \qquad M_P[i, x_j x_k] = \begin{cases} P_{x_i x_k} & i = j \\ P_{x_i x_j} & i = k \\ P_{x_i x_j x_k} & \text{else.} \end{cases}$$

Then $V_0^\perp$ *is the left kernel of* M_P *and* $\delta^* = \mathrm{rank}_{\mathbb{F}_2} M_P$*, computable in* $O(h^4)$ *arithmetic from the phase polynomial.*

Proof. $w \in V_0^\perp$ iff $\Phi(c) \in W_{w^\perp}$ (Corollary 10) iff $\varphi(P) = 0$ for every functional with $\hat{\varphi}|_{w^\perp} = 0$ (Lemma 11) iff $\sum_T [\mathrm{ml}(\ell_w q)]_T P_T = 0$ for all $\deg q \le 2$ (Lemma 12). The expression is linear in w for fixed q, so the condition over the q-basis is $wM_P = 0$; rank–nullity finishes. □

Corollary 14 (Constructive representative)**.** *From* $\ker M_P$ *compute* V_0*, express* P *in the basis of Corollary 8 for* W_{V_0} *(a linear solve of size* $f(\delta^*)$*), and obtain an equivalent representation on at most* $f(\delta^*) = O((\delta^*)^3)$ *parities inside* V_0*; the full* $\mathrm{poly} \cdot 2^{\delta^*}$ *pipeline of Theorem 2 is thus polynomial-time preprocessed.*

Two structural remarks. First, the top-degree block of M_P alone is the classical trilinear flattening whose kernel gives the "effective dimension" of a cubic form modulo $\mathrm{RM}(2, m)$ (Brier–Langevin [BL03, KGP26a]); the collapse rules routing pair and linear grades into the

reserved slots are exactly what lifts this classical invariant to the full $\mathbb{Z}_8$ Clifford quotient. Second, the naive approaches fail instructively: every XOR-shift derivative of a parity-phase function is Clifford (a shift flips coefficients $a \mapsto -a$, an even change), so period-group methods see nothing; and the GL action on the quotient is *twisted* ($z_a \mapsto z_a \oplus z_b \oplus z_c$ sends a singleton parity to an XOR parity, which gains cubic terms), so essential-variable algorithms for polynomials under substitution [Car06, Kay11, Mon17] do not apply.

5 The $4k-1$ law for field multipliers

The `feynman` benchmark suite realizes $\mathrm{GF}(2^k)$ multiplication in Mastrovito form: k^2 partial products $a_i b_j$, each a CCZ between two input wires and a Hadamard-opened product wire, grouped by total degree $m = i + j$ into $2k-1$ classes, with modular reduction by CNOTs between closed Hadamard blocks. We verified these structural premises mechanically for every benchmark file ($k \le 16$; measurements consistent through $k = 256$).

Theorem 15 (The $4k-1$ law)**.** *For these circuits, $\delta^* = \rho = 4k-1$.*

Proof. The image of one CCZ in the Clifford quotient is the single monomial xyz (the function $\pm 4xyz$ mod 8 has zero linear and pair grades; $\mathrm{CCZ}^\dagger$ is identical mod 8). Hence

$$P \;=\; T \;:=\; \sum_{i,j \in [0,k)} x_i y_j z_{i+j},$$

the convolution tensor, supported on $U = \mathrm{span}\{x_i\} \oplus \{y_j\} \oplus \{z_m\}$ with $\dim U = 4k-1$. Upper bound: the pattern is its own coset element, so $V_0 \subseteq \mathrm{span}(\mathrm{supp}\, c) \subseteq U$. Lower bound: by Theorem 13, $u = (\alpha, \beta, \gamma) \in V_0^{\perp}$ requires every contraction $D(u)_{JK} = \sum_I u_I T_{IJK}$ to vanish. Taking $\{J, K\} = \{x_i, y_j\}$ gives $\gamma_{i+j} = 0$ for all i, j, so $\gamma = 0$ (every $m \in [0, 2k-2]$ is a sum); $\{x_i, z_m\}$ gives $\beta_{m-i} = 0$, so $\beta = 0$; symmetrically $\alpha = 0$. Hence $V_0 = U$. □

Corollary 16. (a) *$\rho = \delta^*$: no Clifford/null-identity optimization compresses a Mastrovito multiplier below its raw parity span (matching 12/12 exact benchmark measurements, $k = 4$ through 256).* (b) *$t^* \ge \delta^* = 4k-1$: a linear lower bound on multiplier T-count in the coset model.* (c) *Since a bilinear algorithm with R products of linear forms rewrites the invariant as R rank-one trilinear terms on parities of the same variables, a legal coset element with at most $7R$ parities, we get $t^* \le 7\, M_{\mathrm{bil}}(k)$, where $M_{\mathrm{bil}}(k)$ is the bilinear complexity over $\mathbb{F}_2$ of multiplying two polynomials of degree $< k$. This is $O(k^{\log_2 3})$ by Karatsuba. It is $O(k)$ because the product has degree $\le 2k-2$ and is therefore determined by its residue modulo an irreducible polynomial of degree $2k-1$, so $M_{\mathrm{bil}}(k) \le \mu_2(2k-1)$, the bilinear complexity of multiplication in $\mathrm{GF}(2^{2k-1})$, which is linear by Chudnovsky-type constructions [KGP26b, CC88]. Hence $t^*(\text{multiplier}) = \Theta(k)$.*

Remark 17. The proof uses only that each slot-contraction of the tensor pins one coordinate: any "grouped products" structure $x_i y_j z_{\pi(i,j)}$ with π surjective and slot-injective has $\delta^* = \#x + \#y + \#\text{classes}$. Convolution-type arithmetic (polynomial multiplication, carry-free multiplier layers) falls in this class, which is why this family occurs in real benchmark suites rather than only in constructions.

Two consequences limit what the multiplier family shows. First, by Corollary 16(a) the coset minimization gains nothing on multipliers: 2^{4k-1} is the cost of enumerating the raw parity span. Second, by Corollary 16(c) the rank price 2^{4k-1} and the *optimal* weight price $2^{0.396 t^*}$ with $t^* = \Theta(k)$ differ only in the constants of the exponent. If $M_{\mathrm{bil}}(k) \le 1.44k$ then $0.396\, t^* \le 0.396 \cdot 7\, M_{\mathrm{bil}}(k) \le 4k$ and the optimal weight price is no larger than the rank

price up to a constant factor. The converse does not follow, because $7\,M_{\mathrm{bil}}(k)$ is only an upper bound on t^*: at $k = 2$, $M_{\mathrm{bil}} = 3$ and $7\,M_{\mathrm{bil}} = 21$, while the raw pattern already has 18 parities. All known constructions have $M_{\mathrm{bil}}(k) > 1.44k$. The provable superpolynomial separation lives on generic families, next.

6 Separation on generic dense cores

Theorem 18. *For all but a vanishing fraction of the $2^{f(d)}$ Clifford classes of phase functions on d variables, $\delta^* = d$ and $t^* = \Theta(d^2)$; hence the rank price is $2^{\delta^*} = 2^{\Theta(\sqrt{t^*})}$ against an optimal weight price of $2^{0.396t^*}$, a superpolynomial separation in t^*.*

Proof. Full core. By Corollary 10, a class with $\delta^* < d$ has its core inside one of the $2^d - 1$ hyperplanes H, that is $P \in W_H$, and $|W_H| = 2^{f(d-1)}$ by Corollary 8. The fraction of such classes is at most

$$\frac{(2^d - 1)\,2^{f(d-1)}}{2^{f(d)}} \;=\; \frac{2^d - 1}{2^{\,1+(d-1)+\binom{d-1}{2}}} \;<\; 2^{-\binom{d-1}{2}}.$$

(The exact count follows from the unique core: with $g(j)$ the number of full-core classes on j variables, $2^{f(d)} = \sum_j \binom{d}{j}_2 g(j)$; at $d = 4$ this gives 1536 non-full classes of 16384, matching exhaustive enumeration.) Lower bound on t^* by counting: there are $2^{f(d)}$ classes, while representations with t odd parities number at most $2^{t(d+3)}$; so for most classes $t^* \geq f(d)/(d+3) \sim d^2/6$. Upper bound $t^* \leq \binom{d+1}{2} + 1$ by [Van24, Thm. 3]. Measured exact optima (coset scans to 2^{22} codewords): median $t^* = 3, 7, 11.5$ at $d = 4, 5, 6$. □

Remark (scope). On d variables $\delta^* \leq d$ always, so in this family the rank price 2^d equals the cost of enumerating all path variables directly. The theorem separates that cost from the stabilizer-rank bar $2^{0.396t^*}$; it does not show an advantage of coset minimization over direct enumeration. That advantage appears when $\delta^* < \rho$, as on 26 of the 40 benchmark circuits below.

7 Benchmarks and a wall-clock duel

Suite. Over the 45-circuit `feynman .qc` suite (inputs symbolic; Toffoli via $H \cdot \mathrm{CCZ} \cdot H$; δ^* by Theorem 13), 40 circuits computed: the rank price 2^{δ^*} beats the generic weight bar $2^{0.396t}$ on 33/40. The seven losses are `barenco_tof_3`, `barenco_tof_5`, `barenco_tof_10`, `tof_3`, `csum_mux_9`, `mod5_4` and `qft_4`; Table 1 shows a representative slice. The coset optimization of Theorem 13 reduced δ^* strictly below the raw parity rank on 26/40 circuits; with the raw rank ρ in place of δ^* the count of wins is 23/40. We emphasize the bar is the *generic* exponent: structured decomposers (below) can do better. The values are those of the extracted phase polynomial (see the scope remark after Theorem 2).

Duel. Task with closed-form truth: $A = \langle +^{2k}, c_0{=}1 |\, C_k \,| +^{2k}, 0\rangle = (2^k - 1)/4^k$, which counts multiplicative pairs by amplitude. Our engine (exact $\mathbb{Z}[\omega_8]$, constraint substitution, $\rho = 4k - 1$ residual enumeration with exact $\mathbb{Z}_4$ Gauss sums; Theorem 2) against `quizx`'s cat-state decomposer [KvdW21, KvdWV22]; see Table 2. On this family $\delta^* = \rho$, so the duel measures enumeration of the raw parity span.

The T-count-driven ratio *accelerates* (t grows quadratically in k for the fixed realization; the cat driver's effective $\alpha \approx 0.1$ is far below the generic 0.396); ours is a provable flat 16× per k. At equal implementation language (the measured interpreter handicap is $\sim 500\times$

Table 1: Representative benchmark rows (full data in the repository). t is the post-`full_reduce` T-count where `pyzx` parses the circuit, else the raw parity count; ρ the raw odd-parity rank; δ^* by Theorem 13. Rank wins iff $\delta^* < 0.396\,t$. Rows marked † differ from version 1 (Appendix B).

circuit	t	ρ	δ^*	$0.396\,t$	verdict
`gf2^10_mult`	410	39	39	162.4	rank
`gf2^32_mult`	4128	127	127	1634.7	rank
`gf2^256_mult`	262400	1023	1023	103910	rank
`mod_mult_55`	35	12	8	13.9	rank
`adder_8`†	173	72	55	68.5	rank
`qcla_adder_10`	162	60	50	64.2	rank
`ham15-high`†	1565	526	370	619.7	rank
`grover_5`†	216	104	84	85.5	rank
`qft_4`†	67	43	41	26.5	weight
`barenco_tof_10`†	100	50	41	39.6	weight

Table 2: Wall-clock duel on the counting task $A = \langle +^{2k}, c_0{=}1|C_k|+^{2k}, 0\rangle = (2^k-1)/4^k$. Both engines returned identical exact scalars at every point; wrong-bra controls returned exactly 0. The Python column's per-k ratio measured 15.5× against the provable 16×; `quizx`'s measured ratios accelerate $(27, 52, 65\times)$. The Rust port of Theorem 2 wins from $k = 5$ and reaches $k = 9$; the T-count-driven method's own measured curve prices $k = 8$ at $\gtrsim 74$ hours.

k	`quizx` (Rust, 4 thr)	ours (Python)	ours (Rust)	exact answer
4	0.03 s	2.0 s	0.002 s	15/256
5	0.88 s	48 s	0.028 s	31/1024
6	45.5 s	2119 s	0.63 s	63/4096
7	2970 s	32874 s	29.1 s	127/16384
8	(unreached; $\gtrsim$ 74 h extrap.)	–	460 s	255/65536
9	(unreached)	–	7910 s	511/262144

per state) the crossover is $k \approx 5$; Python-versus-Rust it is $k \approx 9$. A robustness probe (re-expressing each CCZ as seven T gadgets, same unitary) slowed `quizx` only 3–6× (its simplifier re-fuses the gadgets), while δ^* is unchanged, since the re-expression stays inside the coset. Independent of constants, δ^* *forecasts* the cost of the enumeration method in polynomial time before anything runs; decomposition heuristics provide no forecast.

8 Related work

Path-sum representations and Clifford elimination are classical [DHH+05, dBDKR08, Amy19]; "reduce, then enumerate the residual" appears in [Amy19] and, at the mod-2 level with a variable-minimizing linear map, as Montanaro's Corollary 15 [Mon17]; the twisted Clifford quotient is what separates δ^* from that line (Section 4). The weight objective and its Reed–Muller decoding view are due to Amy–Mosca [AM19], with TODD [HC19], TOHPE/FastTODD [Van24], spider nests [dBBW20], and phase teleportation [KvdW20] all minimizing counts; exact T-count is NP-hard [vdWA23]. Khoruzhii, Gelß

and Pokutta [KGP26b] minimize count-type tensor ranks (CP/Waring) of the same cubic invariant on the same multiplier family; theirs is the closest program to ours, and the source of the bilinear coset moves in Corollary 16(c). Brier–Langevin's effective dimension of cubic forms [BL03, KGP26a] preempts the top-grade layer of Theorem 13. Simulation costs shaped like $2^{\mathrm{GF}(2)\text{-dimension}}$ appear as circuit-trace quantities in CAMPS [LC26] and Clifft [CL26a], without coset minimization. Stabilizer-rank simulation and its exponents are in [BG16, BBC+19, QPG21, KvdW22, KvdWV22]; hardness context in [vdWA23, HNS20, JVdN14, Mon17]; mod-8 cubic phases in higher-order Fourier analysis in [Lab22]. Our literature search found no prior formulation of the rank objective, the canonical core, the polynomial-time coset-rank algorithm, or the $4k-1$ law.

9 Open problems

(1) Close the multiplier window: is $0.396\,t^* \geqslant 4k-1$ at the coset optimum? Either answer is a sharp theorem. (2) δ^* versus focused rank-width [CL26b]: which dominates? (3) The doubled (ket/bra) invariant for probabilities and sampling: does structure cancel below $2\delta^*$? (4) Level 4 of the hierarchy ($\sqrt{T}$): the grading becomes quartic; does the syndrome/core machinery lift? (5) An operator-level invariant: extend the minimization to path-sum rewrites that change the number of variables (the Hadamard rule, nonlinear substitution), and prove a lower bound on simulation cost.

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A Machine-verification manifest

All code at `github.com/0verflowme/rankprice` (Python 3.9, standard library; exact arithmetic throughout; `pyzx` 0.10 and `quizx` for cross-checks).

- **Grading iso**: exhaustive at $h=3$ (8^7 vectors; $8^3 \cdot 4^3 \cdot 2$ classes; kernel constant on the grading). Null identities and constructive relation lifts: $h \leq 6$.
- **Lemmas 4–5**: coefficient-vs-count identity on random patterns $h \leq 6$; the vectors $Q(y)$ span a space of dimension $f(h)$, so $\dim\ker\Phi = \dim \mathrm{RM}(h-4,h)^*$, for $h=4,\ldots,7$; random codewords have zero syndrome.
- **Proposition 3**: $t^*(\mathrm{CCZ}) = 7$ and $\delta^*(\mathrm{CCZ}) = 3$ by coset scans at $h=4,5,6$ (2^{22} codewords at $h=6$); syndromes of the 7- and 4-parity patterns differ.
- **Scope remark**: $\delta^* = 2$ for $T\,H\,H\,T^\dagger$ and $\delta^* = 3$ for Toffoli$\cdot$Toffoli; both unitaries equal the identity numerically.
- **Theorem 6**: 45 random configurations; the maximal-monomial witness evaluated to 1 in every case. **Lemma 7**: all 4- and 5-fold reductions verified. **Theorem 9**: 60/60 exact-dimension checks ($h=5,6$).

- **Theorem 13**: 142/142 agreement with brute-force coset scans (120 random; 20 planted-and-scrambled; 2 exhaustive 4.2M-codeword scans at $h = 6$); scaling to $h = 128$, 17,232 gadgets. The rotation into span coordinates preserves δ^* on 240 random patterns.
- **Theorem 15**: premises verified per benchmark file ($k \leq 16$); idealized model exact for $k = 3, \ldots, 12$; $\delta^* = 4k - 1$ measured for all twelve $k \in \{4, \ldots, 10, 16, 32, 64, 128, 256\}$. **Corollary 16(c)**: at $k = 2$ the raw and Karatsuba patterns (18 and 21 parities) have the same syndrome.
- **Theorem 18**: exhaustive count at $d = 4$ (1536/16384 non-full) equals the recursion; samples at $d = 5, 6$ agree.
- **Theorem 2 / Section 7**: amplitudes equal brute force in exact $\mathbb{Z}[\omega_8]$; both engines agree with the closed form at every k; wrong-bra controls exactly 0; Gauss-sum eliminator validated on 320 random forms. Benchmark table recomputed from the `.qc` files.

The checks added in version 2 are in `research/review_checks.py`.

B Changes from version 1

1. **Benchmark table.** A coordinate bug in the benchmark script (the rotation of masks into span coordinates was not linear) overstated δ^* on 24 of the 40 computed circuits. Corrected counts: $\delta^* < \rho$ on 26/40 (was 13/40); rank beats $0.396\,t$ on 33/40 (was 25/40). All multiplier rows are unchanged. Theorems and the 142/142 validation did not depend on the bug.
2. **Proposition 3.** Version 1 cited Selinger's $7 \to 4$ reduction as lowering the weight of the CCZ pattern inside its coset. That gate is $\mathrm{CCZ} \cdot \mathrm{CS}^\dagger$ and lies in a different coset; $t^*(\mathrm{CCZ}) = 7$. The CCZ pattern itself now witnesses the left end.
3. **Scope of the invariant.** "Operator-invariant price" is replaced by invariance under coset moves on fixed path variables, with two identity circuits as examples. Theorem 2 is stated as an upper bound and its proof is written out.
4. **Theorem 18.** The claim that generic classes have full core now has a counting proof and an exact count. A scope remark notes that $2^{\delta^*} = 2^d$ equals direct enumeration in this family.
5. **Corollary 16(c).** The reduction from polynomial multiplication to field multiplication in $\mathrm{GF}(2^{2k-1})$ is stated. The criterion "rank wins iff $M_{\mathrm{bil}}(k) > 1.44k$" is corrected to a one-way implication.
6. **Minor.** The verification of $\ker \Phi$ at $h = 5$ is replaced by a rank computation; open problem (5) is replaced, since the Rust implementation exists (Table 2); the description of the literature search is shortened.

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